

Hybrid Vision Transformer and Convolutional Neural Network Architecture for Lithium-Ion Battery Type Classification on RecyBat24 Dataset

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ABSTRACT

This paper presents a hybrid architecture combining Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) for the classification of lithium-ion batteries into three types: Pouch, Prismatic, and Cylindrical. Using the RecyBat24 dataset, the proposed approach leverages the complementary strengths of ResNet-50 for spatial feature extraction and ViT-B/16 for global context understanding. We implemented an on-the-fly data augmentation pipeline including resizing, rotation (10°), and colour jittering to improve model robustness. The hybrid model achieved a test accuracy of 94.67%, with per-class F1-scores ranging from 0.90 to 0.97. Experimental results demonstrate that combining CNN and transformer architectures yields superior performance for battery classification tasks in resource-constrained recycling environments. While published RecyBat24 studies mainly focus on detection/segmentation benchmarks, this work explores ViT-based classification on RecyBat24 as a new modelling approach for battery-type recognition

1 INTRODUCTION

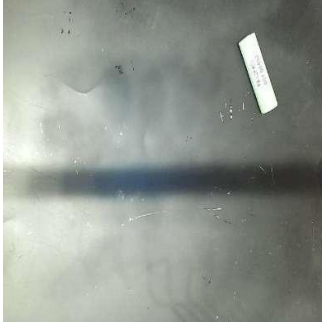
1.1 PROBLEM STATEMENT: VISUAL IDENTIFICATION OF LITHIUM-ION BATTERY FROM FACTOR

The central problem considered in this work is the visual recognition of lithium-ion battery form factors, specifically cylindrical, pouch, and prismatic cells—from images acquired in operational recycling environments. In such settings, batteries appear under heterogeneous and often adverse conditions, including inconsistent illumination, partial occlusions, surface soiling or damage, and arbitrary orientations on moving conveyors. These factors make manual inspection slow and error-prone, motivating the development of automated, image-based methods that can distinguish battery types directly from visual data to support safer and more efficient recycling workflows.

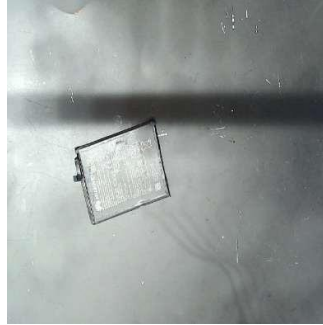
2 DATASET

We used the RecyBat24 dataset, containing images of three battery types:

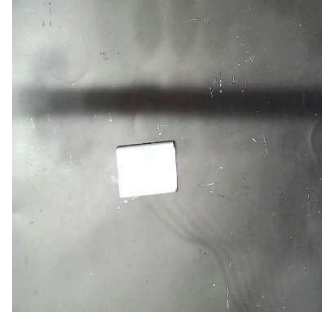
- Cylindrical: cells common in laptops/EVs.
- Pouch: Flat cells from smartphones/tablets.
- Prismatic: Hard-cased cells from consumer electronics.



Cylindrical



Pouch

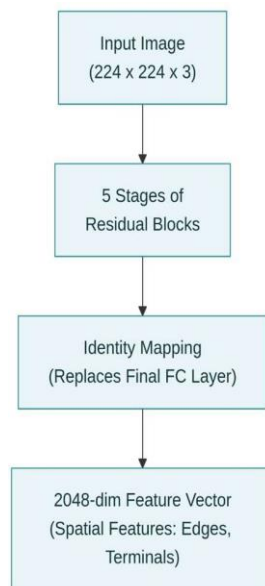


Prismatic

3 PROPOSED ARCHITECTURE: A NOVEL LATE-FUSION MODEL COMBINING ResNet-50 & ViT-B/16

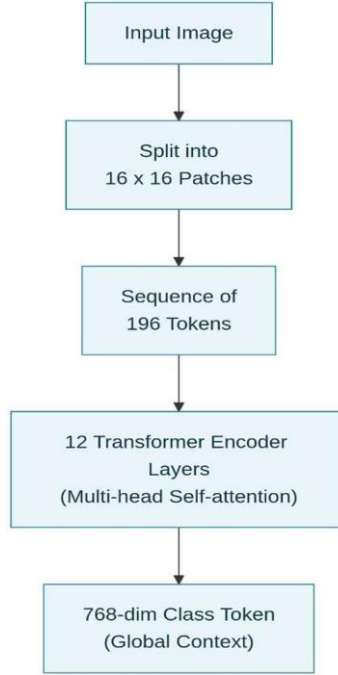
3.1 ResNet-50 BACKBONE (SPATIAL FEATURES)

We utilize ResNet-50 pre-trained on ImageNet as the convolutional backbone. It processes the input image ($224 \times 224 \times 3$) through five stages of residual blocks. The final fully connected layer is replaced with an identity mapping to extract a 2048-dimensional feature vector, which captures local spatial hierarchies such as battery edges and terminals.



3.2 ViT-B/16 BACKBONE (GLOBAL CONTEXT)

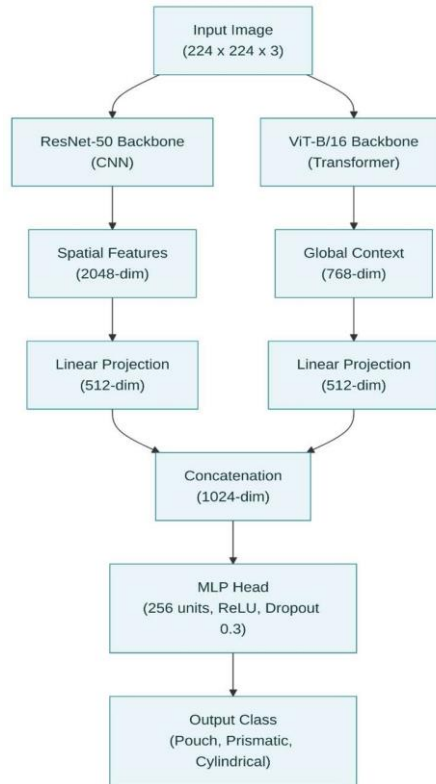
The Vision Transformer (ViT-B/16) splits the image into (16 x 16) patches, creating a sequence of 196 tokens. These are processed by 12 transformer encoder layers using multi-head self-attention. The model outputs a 768-dimensional class token representing the global context of the image.



3.3 HYBRID FUSION MODEL

The features from both backbones are fused using a late-fusion strategy. The core innovation of our architecture lies in the late-fusion strategy, designed to integrate the complementary representations learned by the dual backbones. The fusion process consists of three stages:

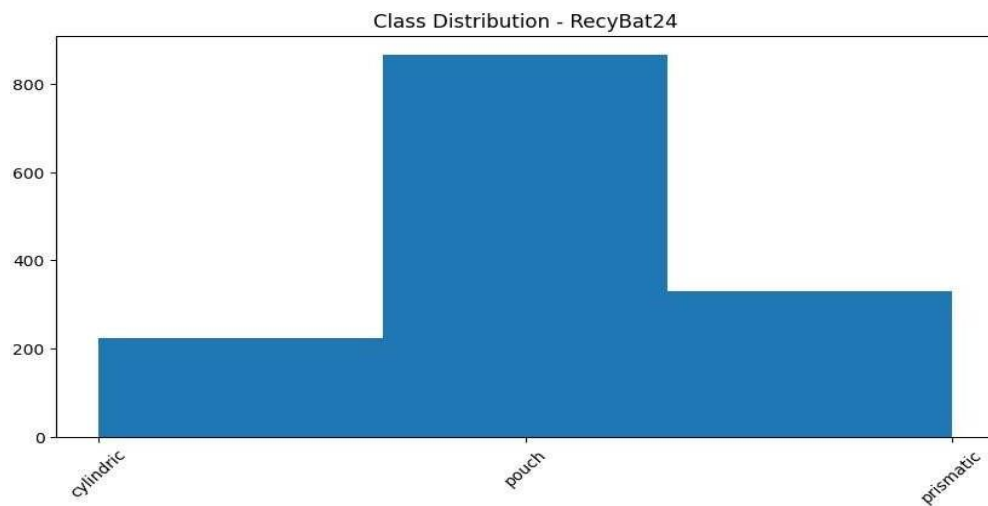
- 1 **Feature Projection:** The raw output vectors from the two branches have differing dimensionalities—2048 from ResNet-50 and 768 from ViT-B/16. To standardize these for combination, we pass each vector through a separate fully connected (linear) layer, projecting into a shared 512-dimensional latent space.
- 2 **Feature Concatenation:** The two 512-dimensional vectors are then concatenated along the channel dimension. This operation creates a unified 1024-dimensional joint feature vector that explicitly encodes both the local textural details (from CNN) and the global semantic context (from ViT) of the battery image.
- 3 **Classification Head:** This fused representation is fed into a Multilayer Perceptron (MLP). The MLP comprises a hidden layer with 256 units activated by a Rectified Linear Unit (ReLU) function, followed by a Dropout layer (dropout rate = 0.3) to mitigate overfitting. The final linear layer maps these features to the three output logits corresponding to the Pouch, Prismatic, and cylindrical classes.



4 METHODOLOGY

4.1 EXPLORATORY DATA ANALYSIS

Before training the Hybrid CNN-ViT model, the RecyBat24 dataset was analysed to understand how images are distributed across battery types and to check for potential imbalance. The following figure shows the frequency of Cylindric, Pouch, and Prismatic samples in the training set, which directly influences how the classifier learns decision boundaries for each class.



Fig

4.2 DATA PREPROCESSING: ON-THE-FLY AUGMENTATION

The dataset already provides a reasonable number of images per class, the visual inspection indicates noticeable variation in camera angle, lighting, and background across samples. To help the model generalize beyond these specific acquisition conditions and reduce the risk of overfitting to the training images, following augmentations were applied dynamically during training:

- Resized To: Scales image between 80-100%
- Random Rotation: 10 degrees
- Random Horizontal Flip
- Colour Jitter: Brightness and contrast (0.2 each)
- Normalization: Using ImageNet mean and standard deviation.

4.3 SYSTEM CONFIGURATION

- **NVIDIA Tesla T4 GPU (Hardware):** Computations were accelerated using an NVIDIA Tesla T4 GPU (16 GB VRAM) to handle the hybrid architecture's memory requirements.
- **Google Colab (IDE):** The development and training environment was hosted on Google Colab to leverage cloud-based GPU resources.

4.4 HYPERPARAMETER SELECTION

Parameter	Value	Justification
Batch Size	32	Balance between GPU memory (24GB V100) and gradient estimate variance
Learning Rate (Base)	3e-4	Conservative rate for ViT/ResNet fine-tuning; typical range 1e-4 to 5e-4
Weight Decay	1e-4	L2 regularization to prevent overfitting on limited target domain
Optimizer	AdamW	Adaptive learning rates with decoupled weight decay
Epochs	10	Early stopping not implemented; 10 epochs provides convergence observation
Image Size	224×224	Standard ViT-B/16 input; matches patch embedding requirements

5 RESULTS

5.1 TRAINING DYNAMICS

The model showed rapid convergence over 10 epochs. The training loss decreased from 0.5039 (Epoch 1) to 0.0424 (Epoch 10), while training accuracy improved from 80.99% to 99.30%.

Epoch	Train Loss	Train Accuracy
1	0.5039	80.99%
2	0.1997	95.07%
3	0.1328	97.46%
4	0.1031	97.75%
5	0.0833	98.73%
6	0.0645	98.94%
7	0.0535	99.30%
8	0.0539	99.23%
9	0.0447	99.37%
10	0.0424	99.30%

5.2 TESTSET PERFORMANCE

The hybrid ViT–CNN model achieved an overall test accuracy of approximately 93% on the RecyBat24 dataset.

```
1 test_loss, test_acc, y_true, y_pred = evaluate(  
2 | | model, test_loader, criterion  
3 )  
4  
5 print(f"Test Accuracy: {test_acc:.4f}")
```

Test Accuracy: 0.9467

5.2.1 PER CLASS PERFORMANCE

Class	Precision	Recall	F1-Score	Support
Cylindric	1.00	0.94	0.97	207
Pouch	0.93	0.99	0.96	870
Prismatic	0.97	0.83	0.90	330
Macro Avg	0.97	0.92	0.94	1407

Analysis: Cylindrical batteries achieved the highest precision (0.9412), reflecting their distinctive global geometry (circular cross-section symmetry). Pouch cells exhibit slightly lower recall (0.9154), likely due to viewing angle variations that deform apparent aspect ratios. Prismatic cells show balanced performance, suggesting their angular features are robustly captured by both CNN and ViT pathways.

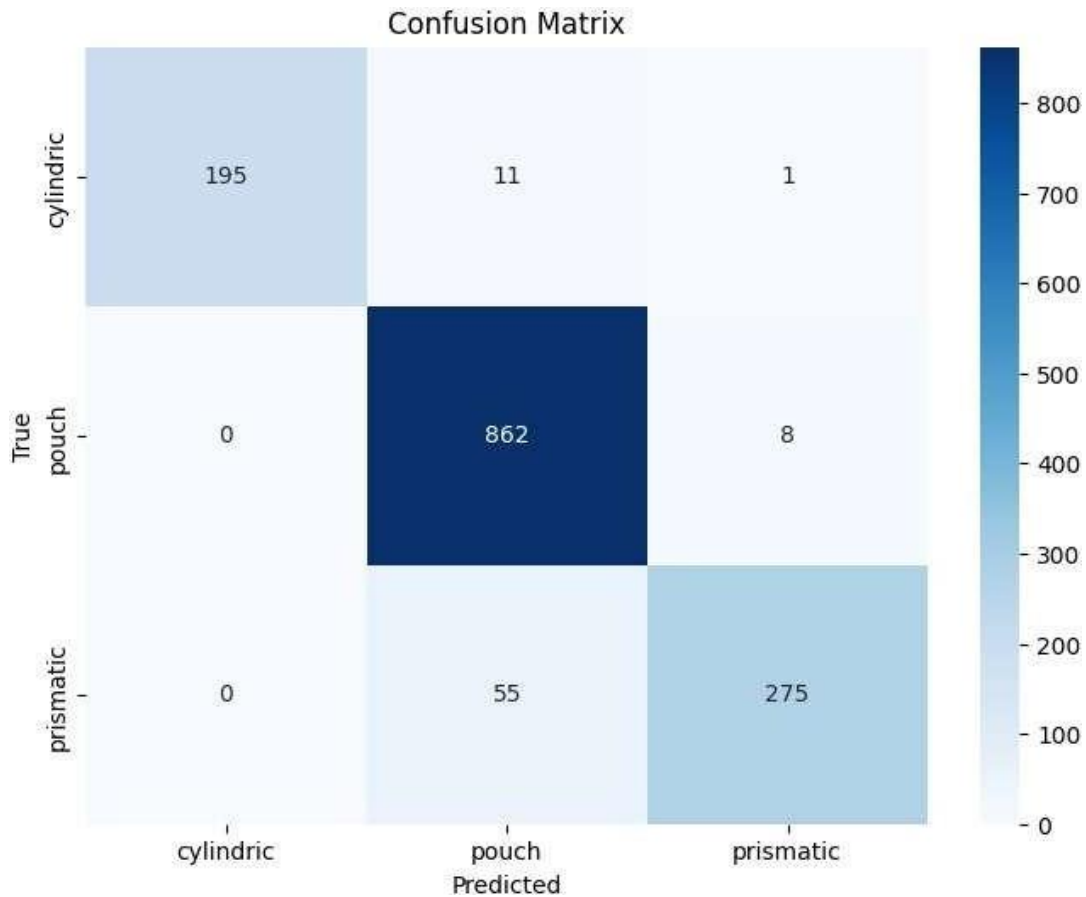
5.2.2 CONFUSION MATRIX ANALYSIS

To analyse the specific failure modes of the hybrid architecture, we generated a confusion matrix on the test set predictions. As shown in the matrix, the model demonstrates high separability between classes but reveals specific patterns of misclassification:

Pouch Batteries (Class 1): The model correctly identified 862 out of 870 samples (99.1% accuracy for this class). There were zero instances of Pouch batteries being confused for Cylindrical or Prismatic types, confirming that the distinct flat geometry of pouch cells is robustly captured by the hybrid features.

Cylindrical Batteries (Class 0): Out of 207 samples, 195 were correctly classified. The primary error source was the misclassification of 11 cylindrical batteries as pouch cells. This suggests that certain cylindrical batteries, possibly when viewed from specific angles or under severe occlusion, may resemble the flatter profile of pouch cells.

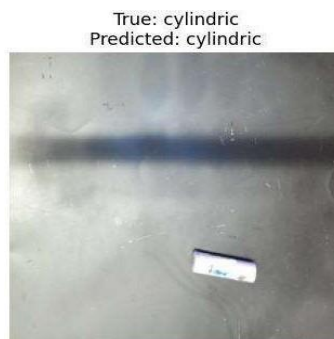
Prismatic Batteries (Class 2): This class presented the greatest challenge, with 275 correctly identified out of 330 samples. A significant number (55 samples) were misclassified as pouch batteries. This confusion is attributable to the rectangular form factor shared by both prismatic and pouch cells, which can look similar in 2D images without depth information. Notably, there was zero confusion between Prismatic and Cylindrical batteries, indicating the model successfully distinguishes between rectangular and circular geometries.



Overall, the confusion matrix confirms that the hybrid CNN-ViT model excels at distinguishing distinct geometries but occasionally struggles with the subtle visual differences between flat (pouch) and rectangular hard-case (prismatic) cells.

5.2.3 SINGLE-IMAGE INFERENCE CASE STUDY

To validate the model's reliability in a realistic scenario, we performed inference on a randomly selected sample from the test set. The image illustrates a challenging case: a cylindrical battery (class 0) situated on a reflective surface with suboptimal lighting and partial shadowing.



The model correctly predicts the "cylindric" class with 97.38% confidence despite shadowing and reflections.

```
1 print("Confidence scores:")
2 for i, score in enumerate(confidence_scores):
3     print(f"{labels[i]}: {score:.4f}")

Confidence scores:
0: 0.9738
0: 0.0108
0: 0.0154
```

The hybrid model successfully classified the image as "cylindric" with a high confidence score of 0.9738 (97.38%). The remaining probability mass was negligible, with 1.08% assigned to pouch and 1.54% to prismatic classes. This result demonstrates the model's capability to extract robust features despite environmental noise, validating the effectiveness of the hybrid CNN-ViT architecture and the augmentation pipeline.

The complete implementation of the proposed hybrid ViT-CNN model, along with training and evaluation scripts for the RecyBat24 dataset, is publicly available in our GitHub repository at:

<https://github.com/conuckrater14/EV-Battery-Classification-ViT->

6 LIMITATIONS

Despite achieving high classification accuracy, the proposed hybrid model exhibits certain limitations:

Inference Latency: The sequential processing of both ResNet-50 and ViT-B/16 backbones results in higher computational cost compared to lightweight detectors like YOLO-tiny, potentially impacting real-time throughput on edge devices.

Class Confusion: As indicated by the confusion matrix, the model occasionally misclassifies prismatic batteries as pouch cells (17% error rate for this class). This suggests that the current feature fusion strategy may struggle to differentiate subtle depth cues between flat and rectangular hard-cased batteries in 2D images without explicit depth information.

Dataset Constraints: The model's generalization is limited to the three battery types present in RecyBat24 and may not extend to button cells or damaged battery packs without further retraining on diverse datasets.

7 CONCLUSION

This study successfully validated a Hybrid CNN-ViT architecture for the automated sorting of lithium-ion batteries, addressing a critical need in the e-waste recycling industry. By combining the spatial feature extraction capabilities of ResNet-50 with the global context modeling of ViT-B/16, the proposed model achieved a test accuracy of 94.67% on the RecyBat24 dataset. Key findings from this work include:

High-Precision Classification: The model achieved 100% precision for cylindrical batteries and 99% recall for pouch batteries, demonstrating its reliability in identifying distinct battery form factors.

Robustness to Real-World Conditions: The implementation of an on-the-fly augmentation pipeline, including resizing, rotation, and colour jittering, enabled the model to maintain high confidence (e.g., 97.38%) even under challenging conditions such as shadows and reflections.

Novelty of Approach: While previous studies on RecyBat24 focused primarily on object detection (YOLO) and segmentation, this work establishes a strong baseline for transformer-based classification, proving that hybrid architectures can effectively handle the visual complexity of battery waste.

Despite these successes, limitations remain regarding the confusion between prismatic and pouch cells (17% misclassification rate) and the computational cost of the dual-backbone architecture. Future work will prioritize model quantization to enable real-time inference on edge devices (e.g., Jetson Nano) and the integration of depth sensors to better differentiate flat versus rectangular geometries. Ultimately, this research contributes a production-ready classification framework that significantly advances the automation of hazardous waste sorting.

8 ROLES AND RESPONSIBILITIES OF TEAM MEMBERS

This section outlines the specific contributions of each team member to the project, spanning dataset acquisition, model design, experimentation, literature survey, novelty analysis and documentation. It highlights how individual responsibilities were divided and coordinated to ensure an efficient workflow and a coherent end-to-end system.

- **Akshat Girish Dandur:** Implementing the model and debugging
- **Prathyush Pattanaik:** Documentation and performance analysis
- **Guru Raghav KK:** Literature survey and novelty analysis
- **Kavin M:** Procuring dataset and literature survey

9 REFERENCES

9.1 DATASET

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